

Compounds

When elements chemically combine in a fixed ratio, they form a compound with brand-new properties — like water from hydrogen and oxygen.

Elements joined chemically

A **compound** is a pure substance formed when **two or more elements combine chemically in a fixed proportion**. Its properties are **different** from those of its elements, and it can be split only by chemical means.

Water (Activity 8.3): passing electricity through (slightly acidified) water breaks it into **hydrogen** (burns with a pop) and **oxygen** (makes a flame glow brighter), in about a 2:1 volume ratio. So water is a compound of hydrogen and oxygen.

Mixture vs Compound

Mixture	Compound
Components just mixed	Elements chemically combined
No fixed ratio	Fixed ratio
Keeps components' properties	New properties
Separated by physical methods	Separated only by chemical methods
No energy change usually	Often heat/light given out or taken in

Classic example: iron + sulfur as a **mixture** can be separated with a magnet; heated to form **iron sulfide** (a compound), the magnet no longer separates them — new properties.

Where students lose marks

Trap 1 — Compound has NEW properties. It does not behave like its elements (water doesn't burn like hydrogen).

Trap 2 — Fixed ratio for compounds, any ratio for mixtures. A compound's elements are combined in a fixed proportion.

Trap 3 — Compounds split only chemically. You can't separate water into H and O by filtering — you need a chemical/electrical process.

Pop test = hydrogen; brighter flame = oxygen.

Model answers — write them like this

Example A. How is a compound different from a mixture? Give one point.

Step 1: In a compound, elements are chemically combined in a fixed ratio with new properties.

Step 2: In a mixture, components are just mixed in any ratio and keep their properties.

∴ Compound = chemically combined, fixed ratio, new properties; mixture = not combined.

Example B. In the electrolysis of water, how do you identify the two gases?

Step 1: Bring a burning candle near each test tube.

Step 2: A pop sound → hydrogen; a brighter flame → oxygen.

∴ Pop = hydrogen, brighter flame = oxygen.

[Ready to practise?](#) → Open the worksheet